

BANGLADESH ARMY UNIVERSITY OF SCIENCE AND TECHNOLOGY, SAIDPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE 3102

Software Engineering Sessional

Project Requirement Analysis Report

Project Name	FarmLink
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Level	3
Term	1
Section	A
Submission Date	

1. Introduction

1.1 Purpose

This report presents the requirement analysis for FarmLink, a proposed system designed to connect surplus, safely edible food from university student halls with nearby livestock farms. The purpose of this document is to record the data, constraints, and stakeholder needs gathered during field visits, and to translate them into a structured set of functional and non-functional requirements for the software system to be developed for this sessional lab project.

1.2 Project Background

University dining halls routinely prepare more food than is consumed, and this surplus is currently discarded as general waste. At the same time, nearby dairy and livestock farms purchase feed daily and have expressed interest in receiving surplus cooked food as a supplementary, cost-saving feed source. FarmLink is intended to act as the digital bridge between these two groups dining hall administrators and farm owners while incorporating veterinary/nutrition expert guidance to ensure that only safe, appropriate surplus food is exchanged.

1.3 Scope

- Digitally record and track daily surplus food quantity, type, and availability windows at each participating hall.
- Match and notify registered livestock farms about available surplus food based on location, animal type, and acceptable food categories.
- Enforce food-safety rules (holding time, spoilage indicators, disallowed food types) drawn from veterinary expert input.
- Support an exchange mechanism (cash, milk, eggs, or meat) agreed between halls and farms.
- Provide reporting on waste reduction and participation for university administration.

1.4 Data Collection Methodology

Requirements were gathered through direct field visits to BAUST (Bangladesh Army University of Science and Technology) student halls in Saidpur, Nilphamari, and to livestock farms in the Nilphamari region. Structured paper-based questionnaires were used: a University Hall Details Form for hall staff, and a Surplus Food-to-Livestock Feeding Program Stakeholder Data Collection Form for farm owners and a veterinary/nutrition expert. All completed forms were signed, dated, and photographed on-site as verification evidence; these photographs and scans are attached in Appendix A and B of this report.

2. Stakeholders

Hall Provost / Assistant Provost / Manager	Registers the hall, records daily surplus food data, approves exchange terms.
Dining-hall Food-Service Staff	Reports actual surplus quantities and timing after each meal service.
Farm Owners	Register their farm, animal inventory, and preferences; accept/reject surplus food offers.
Veterinary / Animal-Nutrition Experts (e.g., BLRI)	Define safe food types, quantities, storage, and holding-time rules used by the system to validate matches.
University Administration	Monitors waste-reduction impact and FarmLink participation across halls.
System Administrator	Manages user accounts, hall/farm onboarding, and system configuration.

3. Field Data — Student Halls

Student halls at BAUST, Saidpur, Nilphamari were surveyed. Verification scans of the completed forms are provided in Appendix A.

3.1 Hall 01 — Abbas Uddin Ahmed Hall

Hall ID	01
University / Institution	BAUST
Location	Saidpur, Nilphamari
Hall Type	Male
Hall Provost	Md. Mahbub Rahman
Assistant Provost	Md. Jahangir Alam
Total Number of Students	351
Average Students Eating / Day	269
Meals Prepared / Day	295
Breakfast / Lunch / Dinner per Day	102 / 290 / 275
Number of Food-Service Staff	14
Common Food Types	Cooked rice, dal, vegetables, fish, chicken, meat, khichuri, mashed food
Meal Producing Most Surplus	Lunch and Dinner
Average Surplus Food / Day	12–18 kg (range observed: 5–25 kg)
Average Surplus Days / Week	6–7 days
Usual Surplus Availability Time	3:20 PM and 10:00 PM
Surplus Holding Time Before Disposal	2–4 hours
Surplus Separated from Other Waste	Yes
Currently Given to Animals	No
Current Disposal Method	General waste bin
Interested in FarmLink Participation	Yes
Preferred Exchange Method	Milk, Eggs, Meat, Cash
Respondent	Md. Jahangir Alam, Lecturer (Civil), dated 31.08.2026

3.2 Hall 02 — Shaheed Dr. Zikrul Haque Hall

Hall ID	02
University / Institution	BAUST
Location	Saidpur, Nilphamari
Hall Type	Male
Hall Provost	Dr. Md. Anower Hossain
Assistant Provost	Masud Parves
Hall Manager	Al Hasen
Total Number of Students	277
Average Students Eating / Day	220
Meals Prepared / Day	660
Breakfast / Lunch / Dinner per Day	93 / 220 / 220
Number of Food-Service Staff	8
Common Food Types	Cooked rice, dal, vegetables, fish, chicken, meat, khichuri, mashed food
Average Surplus Food / Day	12 kg (range observed: 5–25 kg)
Average Surplus Days / Week	6 days
Usual Surplus Availability Time	3:00 PM and 10:00 PM
Surplus Holding Time Before Disposal	2 hours
Surplus Separated from Other Waste	Yes
Currently Given to Animals	No
Current Disposal Method	General waste bin
Interested in FarmLink Participation	Yes
Preferred Exchange Method	Milk, Eggs, Meat, Cash
Respondent	Masud Parves, Lecturer (ME), dated 31.08.2026

3.3 Key Observations from Hall Data

- Student halls report surplus almost every day (6–7 days/week), with a consistent 12–25 kg range — this defines expected daily transaction volume.
- Surplus is consistently available at two fixed windows - afternoon 3 PM and night ~10 PM, suggesting the system should support scheduled rather than purely ad-hoc notifications.
- Surplus is already kept separate from other waste, but must be collected within 2–4 hours before it is discarded — this sets a hard time constraint for the matching/notification system.
- Both halls are willing to accept non-cash returns (milk, eggs, meat) as well as cash, so the exchange feature must support multiple settlement types.

4. Field Data — Livestock Farms

Two livestock farms and two veterinary/nutrition experts were surveyed using the Surplus Food-to-Livestock Feeding Program form. Verification scans are provided in Appendix B.

4.1 Farm 1 — Sabuj Bangla Dairy Farm (Owner: Ziaul Haque)

Location	Nayamatpur, Saidpur, Nilphamari
Animals Kept	Cattle, Goats, Sheep
Herd Size	28 cows, 16 goats
Normal Feed Used	Corn, bran, rice (bhutta, bhushi, bhat)
Feed Purchased / Day	Approx. Tk 6,000–20,000 worth
Approx. Feed Cost	Tk 5,000–7,000
Willing to Accept Surplus Food	Yes
Acceptable Surplus Types	Cooked rice, fruit peels/scraps, other "animal-eatable" food
Unacceptable Types	Cooked curry
Willing Collection Distance	2–5 km
Preferred Return	Cash payment

4.2 Farm 2 — Owner: Hosan (Pirgang)

Location	Pirgang, Nilphamari
Animals Kept	Cattle, Goats, Poultry
Herd Size	20 cattle (7 male, 13 female), 13 goats, 150 hens
Normal Feed Used	Grass, fodder, rice
Feed Purchased / Day	50 kg
Approx. Feed Cost	Tk 7,000–8,000
Willing to Accept Surplus Food	Yes
Acceptable Surplus Types	Cooked rice, cooked vegetables
Unacceptable Types	Mixed food
Willing Collection Distance	2–5 km
Preferred Return	Milk, Eggs

4.3 Expert Input 1 — BLRI (Bangladesh Livestock Research Institute)

Safe food for cattle	Plain cooked rice, vegetable scraps
Safe food for goats	Cooked rice
Safe food for poultry	Cooked rice, vegetables
Safe food for fish	Cooked rice, small vegetable parts
Never give to livestock	Spicy curries, moldy/spoiled food
Factors determining safe quantity	Animal weight/size, species, existing diet balance, nutrient content of food
Signs of contaminated/spoiled food	Sour odor, mold, sliminess
Required storage condition	Cool / dry storage
Maximum safe holding time	2–6 hours
Recommended precaution	Serve within 6 hours

4.4 Expert Input 2 — Independent Veterinary Expert (B.Sc. in Vet.)

Safe food for cattle	Vegetable scraps, crop residues
Safe food for goats	Surplus forage, vegetable peelings
Safe food for poultry	Cooked grains, protein scraps
Safe food for fish	Soft vegetable waste, rice bran
Never give to livestock	Spoiled food, toxic items
Factors determining safe quantity	Animal age, existing diet balance
Signs of contaminated/spoiled food	Bad odor, visual clues, pest presence
Required storage condition	Immediate use only
Maximum safe holding time	Under 2 hours
Recommended precaution	Check ingredients; ensure biosecurity

4.5 Key Observations from Farm & Expert Data

- Both farms consistently accept plain cooked rice and vegetables, but reject spicy/curried or mixed dishes the system must let dining halls tag surplus by food composition, not just quantity.
- Expert opinions on maximum safe holding time diverge (2–6 hours vs. under 2 hours) the system should apply the stricter (more conservative) threshold by default to protect animal health, with configurable overrides per expert/region.
- Farms are within a 2–5 km collection radius of the halls, supporting a location-based matching feature rather than a wide-area marketplace.
- Return preferences differ by farm (cash vs. milk/eggs) the exchange module must be configurable per farm-hall pairing.

5. Functional Requirements

FR-1	The system shall allow dining halls to register with basic details (hall ID, name, institution, location, provost/manager contacts, student and meal counts).
FR-2	The system shall allow hall staff to log daily surplus food entries: quantity (kg), food type(s), time available, and expected holding/expiry time.
FR-3	The system shall allow livestock farms to register with owner details, location, animal inventory, and accepted/unaccepted food types.
FR-4	The system shall store veterinary/expert-defined safety rules per animal type (safe foods, disallowed foods, maximum holding time, storage requirements).
FR-5	The system shall automatically match surplus food entries to nearby registered farms based on distance, animal type compatibility, and food-safety rules.
FR-6	The system shall notify matched farms of an available surplus food offer, including quantity, type, and pickup deadline before it becomes unsafe.
FR-7	The system shall allow a farm to accept or decline an offer, and record the selected exchange method (cash, milk, eggs, or meat).
FR-8	The system shall prevent a surplus offer from being displayed or matched once it exceeds the maximum safe holding time for the relevant animal type.
FR-9	The system shall log completed exchanges (hall, farm, date, quantity, exchange type) for reporting purposes.
FR-10	The system shall generate summary reports on surplus food reduction and FarmLink participation per hall over a selected time period.
FR-11	The system shall allow administrators to add, edit, or deactivate hall, farm, and expert-rule records.

6. Non-Functional Requirements

NFR-1	Availability: The notification feature must reliably deliver alerts within a few minutes of a surplus entry, given the narrow 2–4 hour safe collection window.
NFR-2	Usability: Data-entry forms for hall staff and farm owners must be simple enough for use by non-technical staff, mirroring the paper forms already familiar to respondents.
NFR-3	Reliability: Food-safety rule checks (FR-8) must be enforced consistently and must default to the more conservative expert threshold when multiple expert opinions differ.
NFR-4	Scalability: The system must support additional halls, farms, and experts beyond the two of each surveyed in this initial phase.

NFR-5	Data integrity: All submitted forms/records should be traceable to a named respondent and date, consistent with the signed paper forms used during data collection.
NFR-6	Localization: The interface should support both English and Bengali, since source data and stakeholders use both languages.
NFR-7	Security: Farm and hall contact information must be accessible only to authorized matched parties and administrators.

7. Constraints and Assumptions

Constraints

- Surplus food must be collected within the hall-specific holding window (2 hours to 4 hours) or it is considered unsafe and reverts to general waste disposal.
- Only food types explicitly marked acceptable by a given farm and consistent with expert-approved food types for that animal species may be matched to that farm.
- Collection distance is currently limited to a 2–5 km radius based on farm owner responses; this is a soft constraint that may be revisited as more farms are onboarded.

Assumptions

- Hall staff will consistently and honestly report surplus quantity and timing, similar to how they completed the paper survey forms.
- Farms have reliable means (phone/app) to receive and respond to notifications within the safe collection window.
- Expert-provided safety rules will be periodically reviewed and updated by qualified veterinary/nutrition personnel.

8. Conclusion

The field data collected from two BAUST dining halls, two livestock farms, and two veterinary/nutrition experts consistently supports the feasibility of the FarmLink concept: dining halls generate regular, predictable surplus food and are willing to participate, while nearby farms are willing to accept appropriately safe surplus food in exchange for cash or in-kind returns. The functional and non-functional requirements outlined in this report particularly around food-type matching, strict holding-time enforcement, and flexible exchange settlement are directly derived from this verified field evidence and will guide the design and implementation phases of this sessional lab project.

Appendix A — University Hall Details Forms (Field Verification)

The following scanned forms and site photographs were collected during the field survey and are provided as verification of the data reported in Section 3.

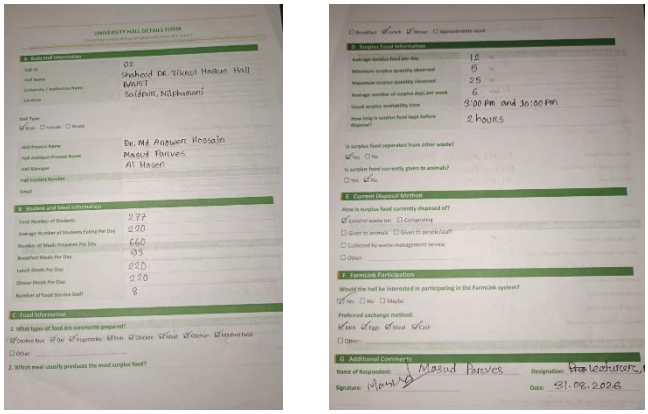


Figure A.1 : Shaheed Dr. Zikrul Haque Hall (Hall 02), Section A & B, C,D Information

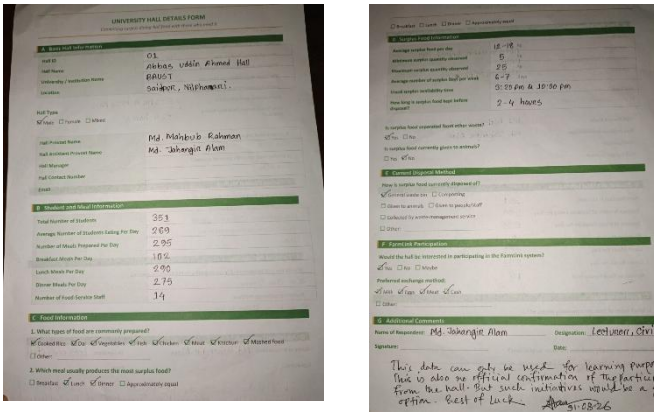
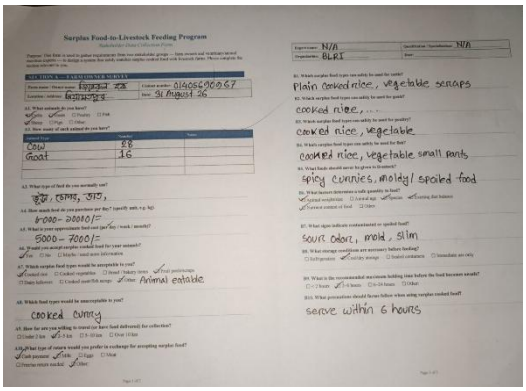
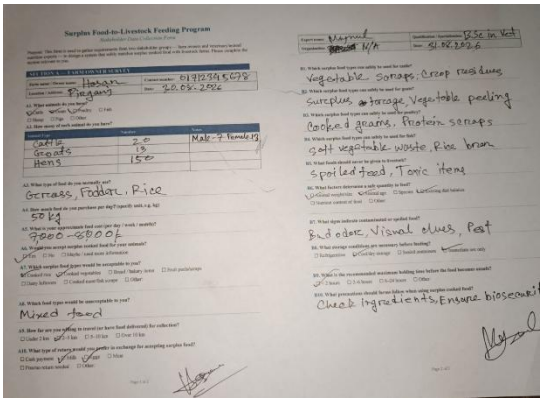


Figure A.2 : Abbas Uddin Hall (Hall 01), Section A & B,C, D Information

Appendix B —Surplus Food to Livestock Feeding Program Forms



B.1 — Farm Owner Survey (Ziaul Haque, Nayamatpur) and BLRI Expert Response



B.2 — Farm Owner Survey (Hosan, Pirgang) and Independent Veterinary

Appendix C — Field Visit Photographs

Photographs taken during the site visits to the dairy farm(s) surveyed, showing the research team, livestock housing, and herd, as further verification of fieldwork.



C.1 — Research team on-site at Sabuj Bangla Dairy Farm signage, Saidpur,



C.2 — Dairy cattle housed at the surveyed farm



C.3 — Close view of cattle in the feeding/holding area



C.4 — Research team member conducting the farm-owner interview



C.5 — Research team with farm owner and staff at the cattle shed